

How To Set Up A Business Unit For CRIMPED-WIRE CONNECTOR MANUFACTURING

Research Team,
ComConnect
Consulting

*This project report is
focussed on setting
up a production unit
for crimped-wire
connectors*

Electronic product packaging has evolved a lot to support the increasing demand for products with complex modular forms of construction. Complex construction needs a wide variety of connectors for inter-connection of different functional units within an electronics equipment/system. Thus, connectors have been gaining importance as a component with advancements in electronic equipment.

Majority of connectors used nowadays has an insertable or removable contact. These are assembled with wires, cables and harnesses, and then inserted into dielectric housing. Crimped-wire connectors are solderless in nature. These are typically used to terminate stranded wires.

There are different types of crimp connectors. But, the most common ones are barrel and open barrel. With barrel-type connectors, a stranded wire is inserted into the cylindrical metal opening (hence, barrel) and then crimped. This is perhaps the most common type of connector.

An open barrel connector is also aptly named, as the barrel is spread open into V or U

Advantages of using crimped-wire connectors

Crimp-type contact provides structural flexibility and brings the following benefits over soldering and wire-wrapped connectors:

1. Easy replacement of damaged contact at assembly stage
2. Easy removal and reinsertion of contact, enabling resolution of wiring errors
3. Uniform and reproducible characteristics of all electrical and mechanical terminations
4. More reliable due to insulation support at contact terminal
5. A well-engineered and well-executed crimp is designed to be gas-tight; prevents oxygen and moisture from reaching the metals (which are often different metals) and, in turn, causing corrosion
6. Mechanically stronger joint, as no alloy used (as in solder)
7. Crimped connections can be used for cables of both small and large cross-sections; whereas, only small cross-section wires can be used with wire-wraps

Business compliances

In the beginning, as a manufacturer, apply for different registrations and licences from the relevant government authorities. However, it is advisable to check the state law.

1. Register the business with Registrar of Companies (ROC).
2. Apply for trade licence from local municipal authority.
3. Apply for MSME Udyog Aadhaar for small- and medium-unit operations. This can be registered online.
4. Obtain NOC from Pollution Control Board. First, apply for Consent to Establish and then for Consent to Operate.
5. Apply for GST registration.
6. Protect brand name with trademark registration.

shape. A wire is cradled into the connector before it is crimped close. Since the wire can be laid into the connector—as opposed to being threaded into it as with a barrel connector—this type of connector is easier to use in automated applications. Connections are generally stronger than those created by barrel connectors. The strength and ease of use make this type of connector common for automotive and industrial uses.

There is much debate over the advantages and disadvantages of crimped and soldered connections. The connection that makes the most sense depends heavily on specific size and type of wire as well as application.

Crimped-wire connectors are mainly used for the interconnection of printed circuit boards (PCBs). Considering huge domestic and export demand, such connectors with good quality and competitive prices have enough market potential.

Manufacturing process

The manufacturing process starts with stamping the crimp from the plated sheet of brass or phosphor bronze using high-speed and high-accuracy stamping press. This involves crimping of the wire terminal. The compressed joint is called a crimp joint,